

Face Recognition Based Smart Attendance System

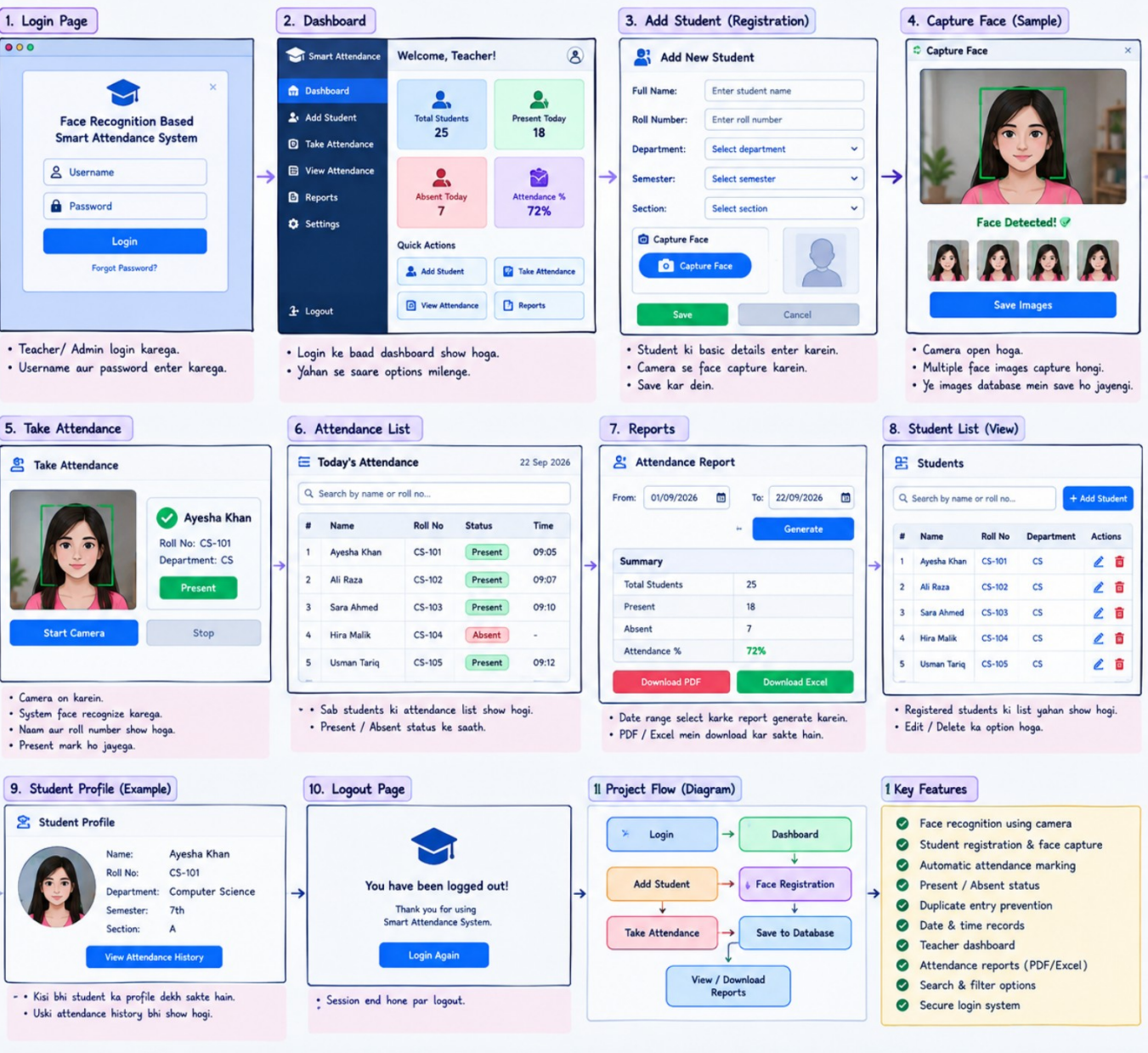
Project Model + Software Requirements Documentation

Project Type	UI / System Model Prototype
Implementation Status	Non-working prototype / structural model
Primary Users	Teacher / Admin
Core Concept	Student registration, face capture, attendance marking, records and reports

This document combines the supplied project model with its supporting documentation in a single file for submission and GitHub upload.

1. Project Model / UI Structure

The following page contains the complete supplied UI model. It represents the proposed structure and flow of the system; it is not presented as a fully working software implementation.



2. Project Name

Face Recognition Based Smart Attendance System

A proposed smart attendance management system that uses face recognition as the main method for identifying registered students and recording attendance. The current project is a UI/model prototype showing the intended screens, workflow, modules and requirements rather than a working implementation.

3. Background

Traditional attendance methods such as manual registers and roll calls can require significant time and may lead to recording mistakes. A smart attendance system can organize student records and attendance information in one place. The proposed system combines student registration, face-image capture, attendance marking, attendance history and reporting into a single interface.

The model is designed as a structured software project for an educational environment. In the proposed workflow, a teacher or administrator logs in, registers students, captures sample face images, takes attendance, views attendance records and generates reports.

4. Objectives

- Design a clear and organized interface for a smart attendance system.
- Provide a central place for student registration and student information.
- Represent a face-capture and face-recognition workflow for attendance.
- Record Present / Absent status along with date and time.
- Allow teachers or administrators to view and search attendance records.
- Generate attendance summaries and reports for a selected date range.
- Provide student profiles and attendance history.
- Reduce repetitive manual attendance work in the proposed workflow.
- Demonstrate a complete project structure that can later be converted into a working system.

5. Stakeholders

Stakeholder	Role / Interest
Teacher / Admin	Logs in, registers students, captures face samples, takes attendance, views records and g
Students	Are registered in the system and have attendance records associated with their profile.
Institute / Department	Uses attendance information for monitoring, record keeping and reporting.
System Developer / Project Team	Designs, implements, tests and maintains the application.

6. Requirements

6.1 Functional Requirements

ID	Requirement	Description
FR-01	Login	The system shall provide a login screen for an authorized Teacher/Admin using username and password.
FR-02	Dashboard	The system shall display a dashboard containing attendance and student summary information and quick actions.
FR-03	Student Registration	The system shall allow the teacher/admin to enter student details such as name, roll number, department, and photo.
FR-04	Face Capture	The system shall provide a camera/capture interface for collecting sample face images for a registered student.
FR-05	Save Registration	The system shall allow student details and captured face samples to be saved to the proposed database.
FR-06	Face Recognition	During attendance, the proposed system shall identify a registered student using a captured face image.
FR-07	Attendance Marking	The system shall record Present/Absent status for students and show the student's name and roll number.
FR-08	Date & Time	The system shall associate attendance records with the relevant date and time.
FR-09	Attendance List	The system shall display attendance records in a searchable list with student name, roll number, status and date.
FR-10	Reports	The system shall allow a date range to be selected and an attendance report to be generated.
FR-11	Export	The system shall provide options to download attendance reports in PDF and Excel formats in the proposed system.
FR-12	Student List	The system shall display registered students and provide Edit/Delete actions.
FR-13	Student Profile	The system shall display a student's profile and attendance history.
FR-14	Search & Filter	The system shall support searching records by student name or roll number and filtering relevant lists.
FR-15	Logout	The system shall provide a logout function to end the current session.

6.2 Non-Functional Requirements

- Usability: The interface should be simple, readable and easy for teachers/admins to navigate.
- Security: Login credentials and student/attendance information should be protected from unauthorized access.
- Performance: The implemented system should process normal attendance operations without unnecessary delay.
- Reliability: Attendance records should be stored consistently and should not be lost during normal operation.
- Maintainability: The software should be organized into manageable modules such as authentication, students, face capture, attendance and reports.
- Scalability: The design should allow more students, classes and attendance records to be added as the system grows.
- Compatibility: The final implementation should work with the selected operating system, camera hardware and database technology.

7. Main System Modules

Module	Purpose
Authentication	Teacher/Admin login and logout.
Dashboard	Overview of student and attendance statistics with quick actions.
Student Management	Add, view, edit and delete registered student information.
Face Registration	Capture and save sample face images for a student.
Attendance	Use the proposed recognition workflow to identify students and record attendance.
Attendance Records	View searchable attendance lists with Present/Absent status and time.
Reports	Generate attendance summaries for a selected date range and export them.
Student Profile	View student information and attendance history.

8. Proposed Project Flow

- Teacher/Admin logs into the system.
- The dashboard provides access to student registration, attendance, records and reports.
- A new student is registered by entering basic details.
- The student's face samples are captured and associated with the registration.
- During attendance, the camera captures a face and the proposed recognition process identifies a registered student.
- The system records the attendance status together with date and time.
- The teacher can review the attendance list, student profiles and attendance history.
- The teacher can generate reports for a selected date range and export them.
- The user logs out after completing the session.

9. Key Features Shown in the Model

- Face recognition using a camera (represented in the prototype).
- Student registration and face capture.
- Automatic attendance marking concept.
- Present / Absent status.
- Duplicate-entry prevention concept.
- Date and time records.
- Teacher dashboard.
- Attendance reports with PDF/Excel export options.
- Search and filter options.
- Secure login system.

10. Project Scope & Limitations

This submission is a model/prototype rather than a working application. The screens demonstrate the proposed user interface, system flow and feature structure. Actual

face-recognition algorithms, camera integration, database connectivity, authentication backend, duplicate-entry logic and PDF/Excel generation would need to be implemented in the development phase.

11. Future Enhancements

- Implement a real face-recognition model and camera integration.
- Connect the UI to a secure database.
- Add role-based access for different types of users.
- Add class/subject-wise attendance.
- Add notifications for low attendance where required.
- Improve privacy controls for stored face images and student information.
- Add backup and recovery functionality.
- Add more detailed analytics and attendance trends.

12. Conclusion

The Face Recognition Based Smart Attendance System is designed as a structured software project for managing student attendance through a proposed face-recognition workflow. The supplied model presents the main screens and system flow from login and student registration to attendance, records, reports and logout. Together, the model and requirements provide a clear foundation for future implementation while keeping the current submission focused on the system's interface and structure.

Submission Note: This single PDF intentionally combines the UI model and its documentation so it can be uploaded as one project document on GitHub.